



(Session-1)

jQuery

jQuery is a JavaScript Library.

jQuery greatly simplifies JavaScript programming.

jQuery is easy to learn.

jQuery

What is jQuery?

jQuery is a lightweight, "write less, do more", JavaScript library.

The purpose of jQuery is to make it much easier to use JavaScript on your website.

jQuery takes a lot of common tasks that require many lines of JavaScript code to accomplish, and wraps them into methods that you can call with a single line of code.

jQuery also simplifies a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

jQuery - AJAX Introduction

AJAX is the art of exchanging data with a server, and updating parts of a web page - without reloading the whole page.

What is AJAX?

AJAX = Asynchronous JavaScript and XML.

In short; AJAX is about loading data in the background and display it on the webpage, without reloading the whole page.

Examples of applications using AJAX: Gmail, Google Maps, Youtube, and Facebook tabs.

jQuery - AJAX load() Method

The jQuery load() method is a simple, but powerful AJAX method. The load() method loads data from a server and puts the returned data into the selected element.

Syntax:

```
$(selector).load(URL,data,callback);
```

The required URL parameter specifies the URL you wish to load. The optional data parameter specifies a set of querystring key/value pairs to send along with the request. The optional callback parameter is the name of a function to be executed after the load() method is completed.

jQuery

Here is the content of our example file: "demo_test.txt":

```
<h2>jQuery and AJAX is FUN!!!</h2>
```

```
<p id="p1">This is some text in a paragraph.</p>
```

The following example loads the content of the file "demo_test.txt" into a specific `<div>` element:

jQuery

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#div1").load("demo_test.txt");
  });
});
</script>
</head>
<body>

<div id="div1"><h2>Let jQuery AJAX Change This Text</h2></div>

<button>Get External Content</button>

</body>
</html>
```

Let jQuery AJAX Change This Text

Get External Content

After Clicking Button

jQuery and AJAX is FUN!

This is some text in a paragraph.

Get External Content

jQuery

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("p").click(function(){
    $(this).hide();
  });
});
</script>
</head>
<body>

<p>If you click on me, I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>

</body>
</html>
```

If you click on me, I will disappear.

Click me away!

Click me too!

jQuery

The jQuery library contains the following features:

- HTML/DOM manipulation
- CSS manipulation
- HTML event methods
- Effects and animations
- AJAX
- Utilities

jQuery

Why jQuery?

There are lots of other JavaScript libraries out there, but jQuery is probably the most popular, and also the most extendable.

Many of the biggest companies on the Web use jQuery, such as:

- Google
- Microsoft
- IBM
- Netflix

jQuery

Adding jQuery to Your Web Pages

There are several ways to start using jQuery on your web site. You can:

- Download the jQuery library from [jQuery.com](https://jquery.com)
- Include jQuery from a CDN, like Google

CDN-Content Delivery Network

Downloading jQuery

There are two versions of jQuery available for downloading:

- Production version - this is for your live website because it has been minified and compressed
- Development version - this is for testing and development (uncompressed and readable code)

Both versions can be downloaded from [jQuery.com](https://jquery.com).

jQuery

The jQuery library is a single JavaScript file, and you reference it with the HTML `<script>` tag (notice that the `<script>` tag should be inside the `<head>` section):

```
<head>  
<script src="jquery-3.5.1.min.js"></script>
```

Tip: Place the downloaded file in the same directory as the pages where you wish to use it.

jQuery

jQuery CDN

If you don't want to download and host jQuery yourself, you can include it from a CDN (Content Delivery Network).

Google is an example of someone who host jQuery:

Google CDN:

```
<head>  
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>  
</head>
```

jQuery

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("p").hide();
  });
});
</script>
</head>
<body>

<h2>This is a heading</h2>

<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

<button>Click me</button>

</body>
</html>
```

This is a heading

This is a paragraph.

This is another paragraph.

Click me

After Clicking Button

This is a heading

Click me



jQuery Syntax

jQuery Syntax

The jQuery syntax is tailor-made for **selecting** HTML elements and performing some **action** on the element(s).

Basic syntax is: **`$(selector).action()`**

- A \$ sign to define/access jQuery
- A (*selector*) to "query (or find)" HTML elements
- A jQuery *action()* to be performed on the element(s)

Examples:

`$(this).hide()` - hides the current element.

`$("p").hide()` - hides all <p> elements.

`$(".test").hide()` - hides all elements with class="test".

© 20: `$("#test").hide()` - hides the element with id="test".

jQuery

The Document Ready Event

You might have noticed that all jQuery methods in our examples, are inside a document ready event:

```
$(document).ready(function(){  
  
    // jQuery methods go here...  
  
});
```

Tip: The jQuery team has also created an even shorter method for the document ready event:

```
$(function(){  
  
    // jQuery methods go here...  
  
});
```

Use the syntax you prefer. We think that the document ready event is easier to understand when reading the code.

by third party.

jQuery Selectors

jQuery Selectors

jQuery selectors allow you to select and manipulate HTML element(s).

jQuery selectors are used to "find" (or select) HTML elements based on their name, id, classes, types, attributes, values of attributes and much more. It's based on the existing [CSS Selectors](#), and in addition, it has some own custom selectors.

All selectors in jQuery start with the dollar sign and parentheses: `$()`.

jQuery Selectors

The element Selector

The jQuery element selector selects elements based on the element name.

You can select all `<p>` elements on a page like this:

```
$("p")
```

jQuery Selectors

Example

When a user clicks on a button, all `<p>` elements will be hidden:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("p").hide();
  });
});
</script>
</head>
<body>

<h2>This is a heading</h2>

<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

<button>Click me to hide paragraphs</button>

</body>
</html>
```

This is a heading

This is a paragraph.

This is another paragraph.

Click me to hide paragraphs

jQuery Selectors

The #id Selector

The jQuery `#id` selector uses the id attribute of an HTML tag to find the specific element.

An id should be unique within a page, so you should use the `#id` selector when you want to find a single, unique element.

To find an element with a specific id, write a hash character, followed by the id of the HTML element:

```
$("#test")
```

jQuery Selectors

Example

When a user clicks on a button, the element with id="test" will be hidden:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#test").hide();
  });
});
</script>
</head>
<body>

<h2>This is a heading</h2>
|
<p>This is a paragraph.</p>
<p id="test">This is another paragraph.</p>

<button>Click me</button>

</body>
</html>
```

This is a heading

This is a paragraph.

This is another paragraph.

Click me

This is a heading

This is a paragraph.

Click me

jQuery Selectors

The .class Selector

The jQuery `.class` selector finds elements with a specific class.

To find elements with a specific class, write a period character, followed by the name of the class:

```
$(".test")
```

jQuery Selectors

Example

When a user clicks on a button, the elements with class="test" will be hidden:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $(".test").hide();
  });
});
</script>
</head>
<body>

<h2 class="test">This is a heading</h2>

<p class="test">This is a paragraph.</p>
<p>This is another paragraph.</p>

<button>Click me</button>

</body>
</html>
```

This is a heading

This is a paragraph.

This is another paragraph.

Click me

This is another paragraph.

Click me

jQuery Selectors

More Examples of jQuery Selectors

Syntax	Description
<code>\$("*")</code>	Selects all elements
<code>\$(this)</code>	Selects the current HTML element
<code>\$("p.intro")</code>	Selects all <code><p></code> elements with <code>class="intro"</code>
<code>\$("p:first")</code>	Selects the first <code><p></code> element
<code>\$("ul li:first")</code>	Selects the first <code></code> element of the first <code></code>
<code>\$("ul li:first-child")</code>	Selects the first <code></code> element of every <code></code>
<code>\$("[href]")</code>	Selects all elements with an <code>href</code> attribute

jQuery Selectors

<code>\$("a[target='_blank']")</code>	Selects all <code><a></code> elements with a target attribute value equal to <code>"_blank"</code>
<code>\$("a[target!='_blank']")</code>	Selects all <code><a></code> elements with a target attribute value NOT equal to <code>"_blank"</code>
<code>\$(":button")</code>	Selects all <code><button></code> elements and <code><input></code> elements of <code>type="button"</code>
<code>\$("tr:even")</code>	Selects all even <code><tr></code> elements
<code>\$("tr:odd")</code>	Selects all odd <code><tr></code> elements

jQuery Event Methods

What are Events?

All the different visitors' actions that a web page can respond to are called events.

An event represents the precise moment when something happens.

Examples:

- moving a mouse over an element
- selecting a radio button
- clicking on an element

The term "**fires/fired**" is often used with events. Example: "The keypress event is fired, the moment you press a key".

jQuery Event Methods

Here are some common DOM events:

Mouse Events	Keyboard Events	Form Events	Document/Window Events
click	keypress	submit	load
dblclick	keydown	change	resize
mouseenter	keyup	focus	scroll
mouseleave		blur	unload

jQuery Event Methods

jQuery Syntax For Event Methods

In jQuery, most DOM events have an equivalent jQuery method.

To assign a click event to all paragraphs on a page, you can do this:

```
$("#p").click();
```

The next step is to define what should happen when the event fires. You must pass a function to the event:

```
$("#p").click(function(){  
    // action goes here!!  
});
```

jQuery Event Methods

Commonly Used jQuery Event Methods

`$(document).ready()`

The `$(document).ready()` method allows us to execute a function when the document is fully loaded. This event is already explained in the [jQuery Syntax](#) chapter.

`click()`

The `click()` method attaches an event handler function to an HTML element.

jQuery Event Methods

The following example says: When a click event fires on a `<p>` element; hide the current `<p>` element:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
    $("p").click(function(){
        $(this).hide();
    });
});
</script>
</head>
<body>

<p>If you click on me, I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>

</body>
</html>
```

If you click on me, I will disappear.

Click me away!

Click me too!

jQuery Event Methods

dblclick()

The `dblclick()` method attaches an event handler function to an HTML element.

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("p").dblclick(function(){
    $(this).hide();
  });
});
</script>
</head>
<body>

<p>If you double-click on me, I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>

</body>
</html>
```

If you double-click on me, I will disappear.

Click me away!

Click me too!

jQuery Event Methods

mouseenter()

The `mouseenter()` method attaches an event handler function to an HTML element.

The function is executed when the mouse pointer enters the HTML element:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("#p1").mouseenter(function(){
    alert("You entered p1!");
  });
});
</script>
</head>
<body>

<p id="p1">Enter this paragraph.</p>

</body>
</html>
```

An embedded page on this page says

You entered p1!

OK

Enter this paragraph.

iQuery Event Methods

mouseleave()

The `mouseleave()` method attaches an event handler function to an HTML element.

The function is executed when the mouse pointer leaves the HTML element:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("#p1").mouseleave(function(){
    alert("Bye! You now leave p1!");
  });
});
</script>
</head>
<body>

<p id="p1">This is a paragraph.</p>

</body>
</html>
```

An embedded page on this page says

Bye! You now leave p1!

OK

This is a paragraph.



jQuery Event Methods

mouseup()

The `mouseup()` method attaches an event handler function to an HTML element.

The function is executed, when the left, middle or right mouse button is released, while the mouse is over the HTML element:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jque
</script>
<script>
$(document).ready(function(){
  $("#p1").mouseup(function(){
    alert("Mouse up over p1!");
  });
});
</script>
</head>
<body>

<p id="p1">This is a paragraph.</p>

</body>
</html>
```

An embedded page on this page says

Mouse up over p1!

OK

This is a paragraph.

jQuery Event Methods

hover()

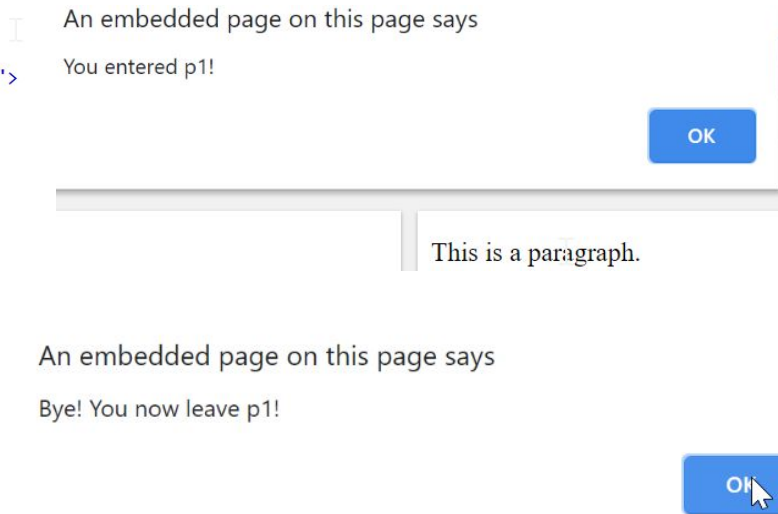
The `hover()` method takes two functions and is a combination of the `mouseenter()` and `mouseleave()` methods.

The first function is executed when the mouse enters the HTML element, and the second function is

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("#p1").hover(function(){
    alert("You entered p1!");
  },
  function(){
    alert("Bye! You now leave p1!");
  });
});
</script>
</head>
<body>

<p id="p1">This is a paragraph.</p>

</body>
</html>
```



jQuery Event Methods

focus()

The `focus()` method attaches an event handler function to an HTML form field.

The function is executed when the form field gets focus:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("input").focus(function(){
    $(this).css("background-color", "yellow");
  });
  $("input").blur(function(){
    $(this).css("background-color", "green");
  });
});
</script>
</head>
<body>
```

Name:

Email:

```
</body>
```

```
</html>
```

blur()

The `blur()` method attaches an event handler function to an HTML form field.

The function is executed when the form field loses focus:

Name:

Email:

Name:

Email:

jQuery Event Methods

The on() Method

The `on()` method attaches one or more event handlers for the selected elements.

Attach a click event to a `<p>` element:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("p").on("click", function(){
    $(this).hide();
  });
});
</script>
</head>
<body>

<p>If you click on me, I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>

</body>
</html>
```

If you click on me, I will disappear.

Click me away!

Click me too!

jQuery Event Methods

Attach multiple event handlers to a `<p>` element:

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
  $("p").on({
    mouseenter: function(){
      $(this).css("background-color", "lightgray");
    },
    mouseleave: function(){
      $(this).css("background-color", "lightblue");
    },
    click: function(){
      $(this).css("background-color", "yellow");
    }
  });
});
</script>
</head>
```

```
<body>
```

```
<p>Click or move the mouse pointer over this paragraph.</p>
```

```
</body>
```

```
</html>
```

Click or move the mouse pointer over this paragraph.

Click or move the mouse pointer over this paragraph.

Click or move the mouse pointer over this paragraph.

Click or move the mouse pointer over this paragraph.

jQuery read and update values

Get & Set Methods in jQuery

These methods are used to get (read) or set (update) values of elements.

1. **.text()**

Get text: `$("#p1").text();`

Set text: `$("#p1").text("New text");`

2. **.html()**

Get HTML: `$("#p1").html();`

Set HTML: `$("#p1").html("<b>Bold Text</b>");`

3. **.val()**

Get value (input, select, textarea): `$("#txtName").val();`

Set value: `$("#txtName").val("Ketan");`

text() Method

```
<p id="p1">Hello <b>World</b></p>
```

```
<script>
```

```
  // Get text
```

```
  alert($("#p1").text()); // Output: Hello World
```

```
  // Set text
```

```
  $("#p1").text("New Text Here");
```

```
  // Now <p id="p1">New Text Here</p>
```

```
</script>
```

html() Method

```
<p id="p2">Hello <b>World</b></p>
```

```
<script>
```

```
  // Get HTML
```

```
  alert($("#p2").html()); // Output: Hello <b>World</b>
```

```
  // Set HTML
```

```
  $("#p2").html("New <i>Italic</i> Text");
```

```
  // Now <p id="p2">New <i>Italic</i> Text</p>
```

```
</script>
```

val() Method

```
<input type="text" id="txtName" value="Ketan">
```

```
<script>
```

```
    // Get value
```

```
    alert($("#txtName").val()); // Output: Ketan
```

```
    // Set value
```

```
    $("#txtName").val("Rahul");
```

```
    // Now input box contains "Rahul"
```

```
</script>
```



Thank you

